

Vibration Control of Long-Span Beams: Experimental and Analytical Study of Beam Structures Incorporated with Connection Dampers

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(Received 2 January 2003; accepted 4 April 2003)

Abstract: In order to reduce the vertical vibrations of long-span beam structures subjected to human activities or machinery running, a type of beam-column connection incorporated with viscoelastic dampers is designed and the effectiveness in vibration control is proved experimentally. An analytical method implemented with a fractional derivative model for structures with such damping devices is then proposed. By comparing analytical results with test data, the method is verified to be able to predict structural behaviors very well. Based on the parametric studies being carried out, useful guidelines are achieved for design of beam structures with connection dampers.

Key Words: Vibration control, viscoelastic dampers, beam-column connections, fractional derivative model

1. INTRODUCTION

Vertical vibrations are always induced by a variety of human activities and undefined office machinery in buildings that have large column-free areas for general use. Excessive floor vibrations have become one of the major problems to be considered in modern building design with the increasing use of high-strength, lightweight structural materials. Moreover, the demand for open-space areas in offices and commercial retail buildings has led to the use of floor systems with longer spans, which results in more flexible structures. Many long-span structures such as gymnasiums and concert halls built in the last few decades have experienced excessive vibrations due to human activities (Bachmann, 1992; Mouring and Ellingwood, 1992). Vibration tests and analysis of a floor system were accomplished by Osborne and Ellis (1995) and it was found that considerable vibration, which was visible by human perception, occurred in long-span structures under human activities. Thus, unfavorable consequences are brought in. To achieve satisfactory dynamic performance, it is very important to find ways to control such vibrations.

Viscoelastic dampers have been used by many researchers in the past years to suppress lateral vibration of building structures under wind and seismic loading. In general, dampers can be installed in various ways according to the deformation behavior of the structures. To account for the story shift of buildings, dampers were mostly installed between adjacent stories with additional bracing or rigid RC-wall (Tsai and Lee, 1993; Chang et al., 1995; Samali and Kwok, 1995; Shen et al., 1995; Cho et al., 1998). Relative movement between beam and column was considered as well. For instance, viscoelastic dampers were incorporated in between the lower chords of the horizontal trusses and the columns of the outside wall in the twin towers of the World Trade Center (Mahmoodi et al., 1987). An energy dissipating connection, in which the elastomeric pads were under tension and compression, was designed and installed in braced frames to resist earthquake excitation by Hsu and Fafitis (1992) and the result of the study showed that the connection could reduce the response of the structure significantly.

If viscoelastic dampers are placed in the beam–column connections properly, effectiveness in vertical vibration control of long-span beams is expected. Hereby, a beam–column connection incorporated with viscoelastic dampers is designed and experimental investigations are carried out on this type of beam–column connection. The fractional derivative model, which has been proved to be able to describe accurately the dynamic behaviors of viscoelastic dampers (Bagley and Torvik, 1983), is employed to develop an analytical method in the time domain. The accuracy of the method is then proved by comparisons of the experimental and analytical results. Afterwards, a large amount of numerical analysis work is performed on a full-scale long-span beam with the designed beam–column connections and some useful guidelines are provided.

2. EXPERIMENTS ON A BEAM WITH VARIOUS CONNECTIONS

2.1. Description of the Beam–Column Connections

A steel beam with two ends connected with very rigid columns is chosen for dynamic tests. The I-section beam, which is 4.54 m long with span-to-depth ratio of 22.7, is connected with columns through angles bolted on both sides of the beam web. The front view of the beam–column connection without the damper is shown in Figure 1(a). The rubber material damper (serial no. HD91) with high viscoelasticity (DHV) manufactured by Wuxi Vibration Isolator Company, China, has been proved to have good capability in energy dissipation (Fang et al., 1996). Two such dampers with dimensions of $125 \times 125 \times 4 \text{ mm}^3$ are proposed for each beam–column connection. Each damper is stuck to two thin steel plates by high-strength glue and then connected to the top or bottom flange of the beam through inner bolts. One cleat angle, which is bolted to the column, is connected on the other side of each damper with inner bolts. The aspect of this type of connection is shown in Figure 1(b). The damper stuck to two plates with inner holes is displayed in Figure 2. When dynamic loading comes, the beam rotates slightly around both ends and the dampers in the connections undergo deformations. In practice, if no damper is used, the beam–column connection is normally configured as in Figure 1(c).

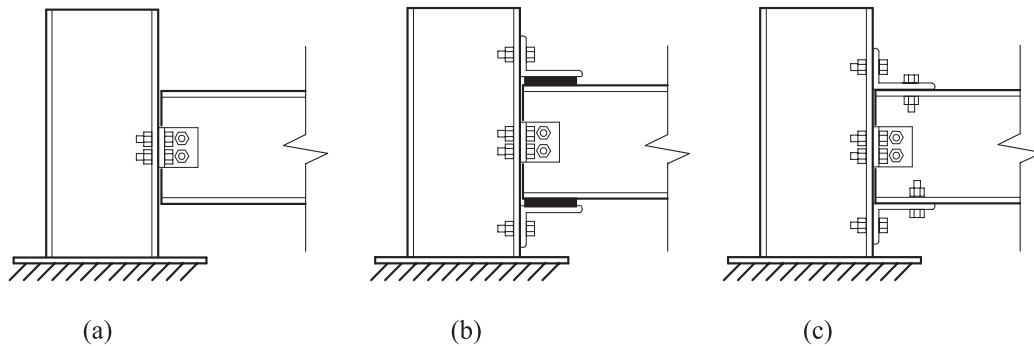


Figure 1. Front view of the beam-column connections: (a) no damper and flange angle; (b) with dampers; (c) with flange angles.

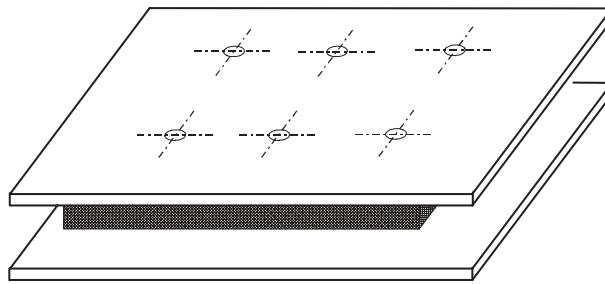


Figure 2. Schematic diagram of the damper stuck to two steel plates.

2.2. Experimental Tests

To measure responses at different positions, seven equally spaced accelerometers are distributed along the beam. The beam is excited by an exciter in the vertical direction at the mid-span. The excitation is dominated by signals (i.e. white noise or sinusoidal wave) generated from a signal generator via a signal amplifier. The force imposed on the beam is monitored by a digital oscilloscope. When the beam is excited, signals collected by the accelerometers and the force transducer are transferred to the Dynamic Test and Analysis System (DTAS) for process. After that, the data are input to a computer and then analyzed by the software for DTAS. The setup of the test system is displayed in Figure 3.

Dynamic tests, called cases I, II, and III, are carried out on the beam with three different beam-column connections as shown in Figures 1(a)–(c), respectively. By applying white noise excitation, the first natural frequency of the beam is identified as 35.2 Hz for case I, 42.3 Hz for case II, and 58.5 Hz for case III. Thus, the frequency steps for tests with sinusoidal loading in different frequency ranges are determined. For excitation frequency far away from the natural frequencies, steps are set very large, i.e. 5–10 Hz. When the excitation frequency is close to the natural frequency, steps would be gradually reduced to 0.05 Hz. At

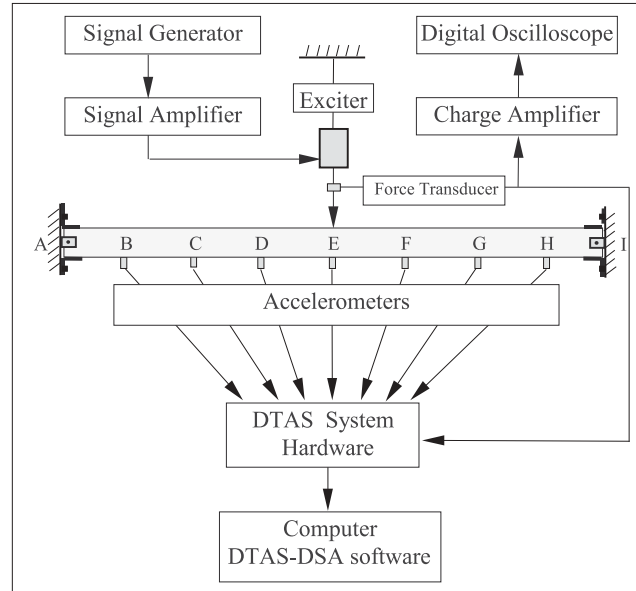


Figure 3. Setup of the test system.

each excitation frequency, the amplitude of the sinusoidal loading is kept at a certain value (i.e. 10 N) with the assistance of the digital oscilloscope. Then the sinusoidal loading and responses at the eight predefined positions are collected by the DTAS. All the experiments are performed at a room temperature of 25°C.

2.3. Effectiveness in Vertical Vibration Control

The acceleration peaks at section E are compared within the first natural frequency regions of the three cases as shown in Figure 4. Obviously, the resonant acceleration of case II is much less than that of cases I and III. The resonant acceleration peaks at section E are 48.33 m s^{-2} for case I, 7.93 m s^{-2} for case II, and 36.26 m s^{-2} for case III. This means that the resonant acceleration peak at section E for case II is only 16.4% of that for case I and 21.9% of that for case III. The damping ratio, another effectiveness indicator, is 0.224% for case I, 1.11% for case II and 0.245% for case III. Figure 5 shows the acceleration peaks at different positions of the beam under excitation at the corresponding resonant frequencies of the three cases; the mode shapes of the beam with different connections are shown. The effectiveness in vibration control for the whole beam is clear from this figure as well.

Experiments within the second and third natural frequency regions are also carried out. As a result, great effectiveness is also achieved by the designed beam-column connections. Because practical excitations are in a low-frequency band (i.e. less than 60 Hz), the study here is focused on the behaviors of the beam in the first natural frequency region.

In addition, the effectiveness in vibration control with the designed beam-column connections has also been proved by dynamic tests on the beam with $100 \times 100 \times 4 \text{ mm}^3$ dampers. The first resonant frequency shifts to 41.7 Hz. The maximum resonant acceleration

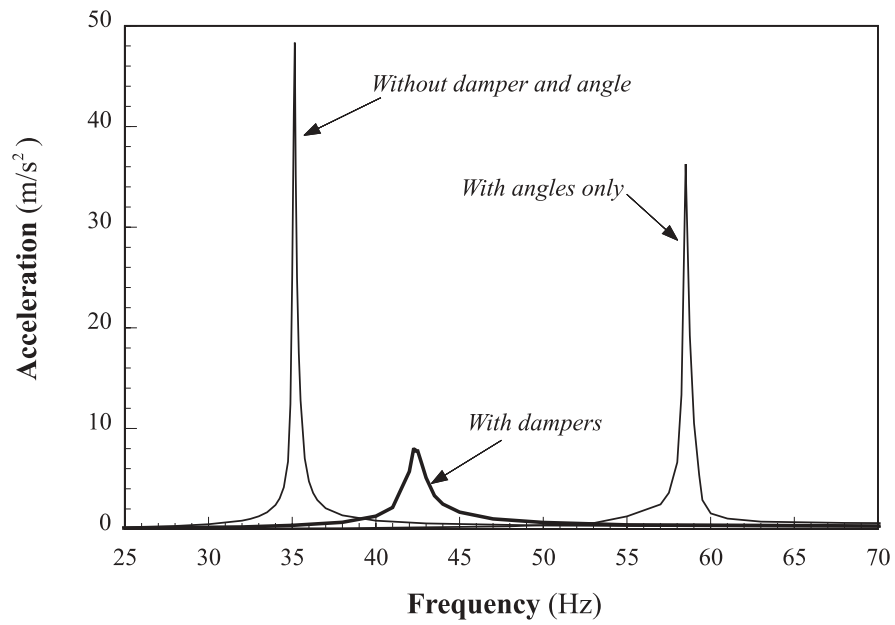


Figure 4. Acceleration peaks at section E of the beam with various beam-column connections.

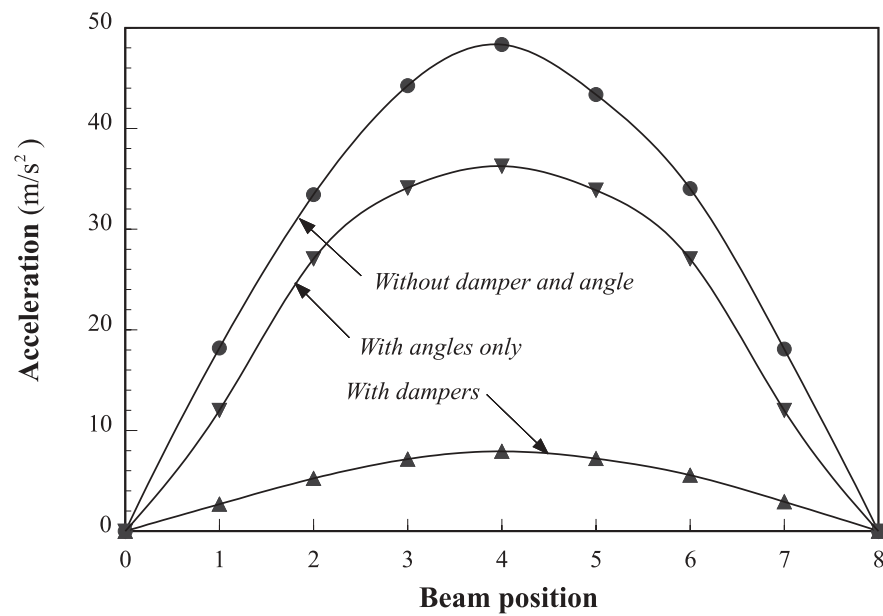


Figure 5. Resonant acceleration peaks at different positions of the beam with various beam-column connections.

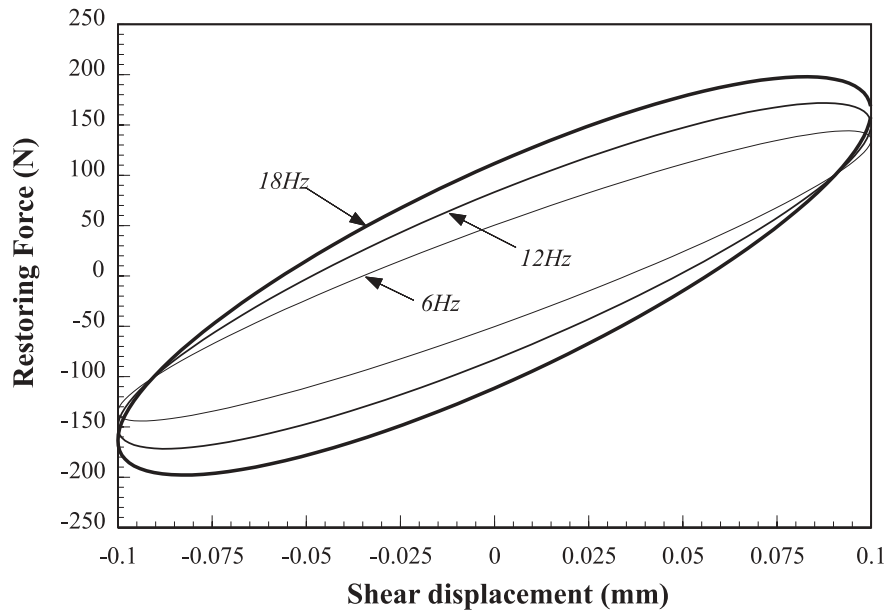


Figure 6. Hysteresis loops of damper DHV due to sinusoidal loading.

is about 9.57 m s^{-2} under the same load conditions and the damping ratio of the beam structure is 0.92%.

3. ANALYSIS FOR STRUCTURES WITH THE DESIGNED CONNECTIONS

3.1. Modeling of the Viscoelastic Damper

Dynamic tests were performed on specimens of DHV by Fang et al. (1996). Several dampers of the same material with various dimensions (i.e. $100 \times 100 \times 4 \text{ mm}^3$, $125 \times 125 \times 4 \text{ mm}^3$, $150 \times 150 \times 4 \text{ mm}^3$ and $100 \times 100 \times 6 \text{ mm}^3$, etc.) have been investigated experimentally in both axial and shear directions at different temperatures with sinusoidal loading. Strain ranges during testing were between 0 and 250% in the shear direction and 0 and 50% in the axial direction. The temperature was increased from 20°C to 35°C . From the study, it has been found that the commonly used generalized derivative models such as the Kelvin–Voigt model cannot well describe the dynamic behavior of DHV, especially the variation of equivalent stiffness with the excitation frequency as shown in Figure 6. As illustrated in Figure 7, the Kelvin–Voigt model can only be used for materials whose equivalent stiffness does not change with excitation frequency. However, the fractional derivative model can overcome this shortcoming. In past years, the fractional derivative model has been found to be more powerful than generalized derivative models (Bagley and Torvik, 1983, 1985, 1986; Koh and Kelly, 1990; Makris and Constantinou, 1991). Because the frequency range is limited for viscoelastic dampers in applications, a three-parameter model is sufficient to describe the dynamic behaviors of the dampers (Bagley et al., 1985). Based on the experimental results

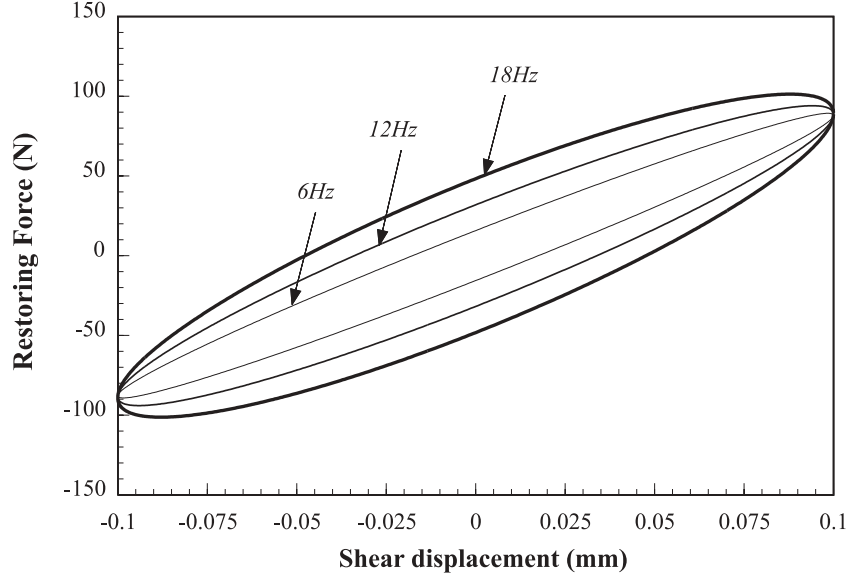


Figure 7. Hysteresis loops described by the Kelvin-Voigt model.

and the research results on the temperature effect by Soong and Dargush (1997), the fractional derivative model is modified to describe the behavior of DHV by the stress-strain relationship in both shear and axial directions, respectively:

$$\tau = \psi(T)(G_0\gamma + G_1D^\alpha\gamma) \quad (0 \leq \alpha \leq 1) \quad (1a)$$

$$\sigma = \psi_a(T)(E_0\varepsilon + E_1D^{\alpha_a}\varepsilon) \quad (0 \leq \alpha_a \leq 1). \quad (1b)$$

Here, D denotes the operator of fractional differentiation, the expression of which will be described later in Section 3.4, τ is shear stress, γ is shear strain, σ is axial stress, and ε is axial strain. G_0 is the shear modulus and E_0 is the axial modulus. G_1 and E_1 are non-unit constants which determine the damping performance. Parameters α and α_a are fractions, and $\psi(T)$ and $\psi_a(T)$ are functions of temperature T in shear and axial directions, respectively.

Based on experimental data collected at the temperature of 25°C, parameters for the selected viscoelastic material have been identified as follows: $\alpha = 0.724$, $G_0 = 0.795 \text{ N mm}^{-1}$, $G_1 = 0.0285$, $\alpha = 0.684$, $E_0 = 3.225 \text{ N mm}^{-1}$ and $E_1 = 0.087$. By studying the parameters identified from different temperatures, it is found that the parameters G_0 , E_0 , G_1 , and E_1 change with temperature in an exponential form, while α and α_a remain unchanged. When temperature increases, parameters G_0 , E_0 , G_1 , and E_1 decrease. The temperature function is evaluated to be $\psi(T) = \psi_a(T) = e^{-0.029(T-25)}$, where T is in Celsius.

Comparisons between the hysteresis loops of test data and the fractional derivative model have proved that the model can well describe the damper behavior under various conditions as experiments (Fang et al., 1996).

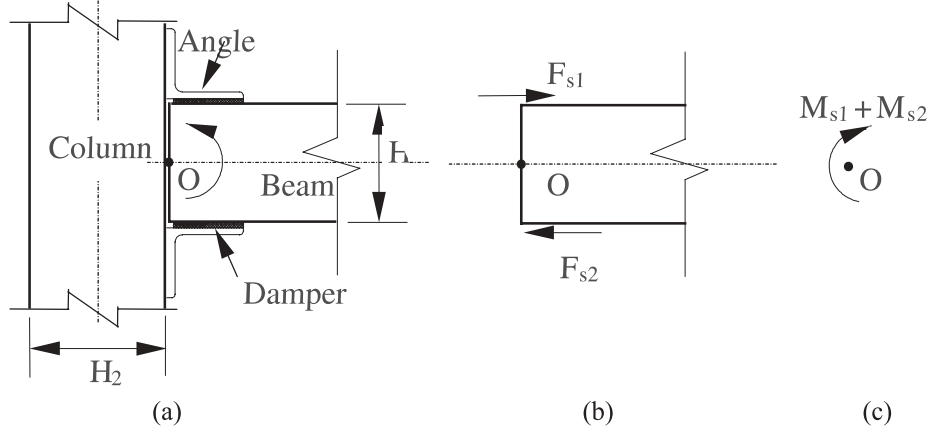


Figure 8. Effect of the shear restoring force of viscoelastic dampers.

3.2. Shear Restoring Forces of Viscoelastic Dampers

As the flexible beam is not fully fixed to the column, it slightly rotates around the columns on both beam ends when it vibrates. Once the beam has a small rotation angle θ around a column, displacement on the flange at the beam end is enlarged by the beam height. Thus, the viscoelastic dampers mainly undergo shear deformation as shown in Figure 8 due to the flange movement. The shear displacement of the damper stuck on the beam flange can be calculated as

$$d = \frac{H \sin \theta}{2} \quad (2)$$

where H is the height of the beam section.

Then the shear restoring force of each viscoelastic damper can be written as

$$F_s = \frac{A_r}{h} \psi(T) (G_0 d + G_1 D^\alpha d) \quad (3)$$

where A_r and h are the shear area and the thickness of the damper, respectively.

With consideration of equation (2),

$$F_s = \frac{A_r H \psi(T)}{2h} [G_0 \sin \theta + G_1 D^\alpha (\sin \theta)] \quad (4)$$

Therefore, the moment about the centroidal axis produced by one damper can be obtained by

$$M_s = F_s \cdot \frac{H}{2} = \frac{H^2 A_r \psi(T)}{4h} [G_0 \sin \theta + G_1 D^\alpha (\sin \theta)] \quad (5)$$

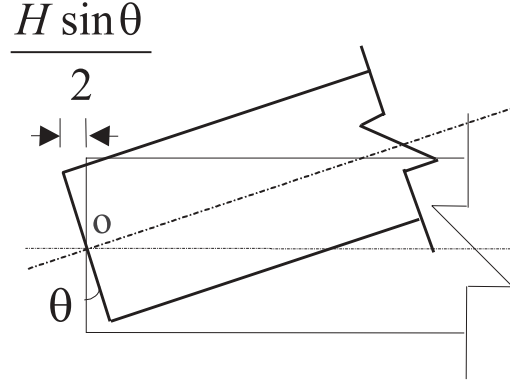


Figure 9. Illustration of beam rotation around point "o".

3.3. Axial Restoring Force of Viscoelastic Dampers

For most conditions, dampers installed at the beam–column connections are subject to axial force as well. When a rotation angle occurs as shown in Figure 9, the damper also undergoes axial deformation. Obviously, the axial deformation changes along the longitudinal direction of the damper. As displayed in Figure 10(a), the left side of the damper on the upper flange is in tension and the right side is in compression. However, the damper on the lower flange is in tension only, which can be seen in Figure 10(b). The deformation in axial direction of the dampers can be expressed as follows:

$$\Delta h(x) = \begin{cases} x \tan \theta - \frac{H}{2}(1 - \cos \theta) & (0 \leq x \leq w) \quad (\text{the upper damper}) \\ x \tan \theta + \frac{H}{2}(1 - \cos \theta) & (0 \leq x \leq w) \quad (\text{the lower damper}) \end{cases} \quad (6)$$

The restoring force in the axial direction of a damper can be denoted as

$$F_a = \int_0^w \sigma(x)b \, dx \quad (7)$$

where w is the total length of the damper.

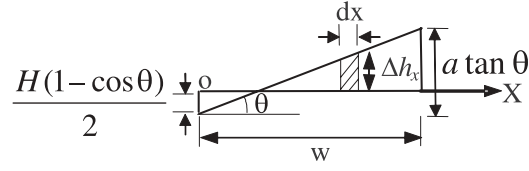
The axial stress can be given by

$$\sigma(x) = \psi_a(T)[E_0 \varepsilon(x) + E_1 D^{\alpha_a} \varepsilon(x)]. \quad (8)$$

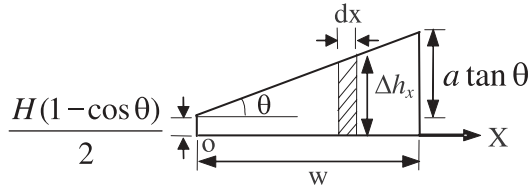
The substitution of the axial strain $\varepsilon(x) = \frac{\Delta h(x)}{h}$ into equation (8) yields

$$\sigma(x) = \frac{\psi_a(T)}{h} [E_0 \Delta h(x) + E_1 D^{\alpha_a} \Delta h(x)]. \quad (9)$$

Thus, the restoring forces of the upper damper and the lower damper can be calculated by equations (10a) and (10b)



(a)



(b)

Figure 10. Distribution of axial deformation of the damper: (a) on the upper flange; (b) on the lower flange.

$$F_a = \psi_a(T) \left\{ \frac{w^2 b}{2h} [E_0 \tan \theta + E_1 D^{\alpha_a}(\tan \theta)] - \frac{wbH}{2h} [E_0(1 - \cos \theta) + E_1 D^{\alpha_a}(\cos \theta)] \right\} \quad (10a)$$

$$F_a = \psi_a(T) \left\{ \frac{w^2 b}{2h} [E_0 \tan \theta + E_1 D^{\alpha_a}(\tan \theta)] + \frac{wbH}{2h} [E_0(1 - \cos \theta) + E_1 D^{\alpha_a}(\cos \theta)] \right\}. \quad (10b)$$

Therefore, the additional moment caused by the axial force can be calculated for the upper and lower damper as equations (11a) and (11b), respectively:

$$\begin{aligned} M_a &= \int_0^w \sigma(x)b(x - \frac{H}{2} \tan \theta) dx \\ &= \frac{w^2 b}{h} \psi_a(T) \left\{ E_0 \left[\frac{w \tan \theta}{3} - \frac{H}{4} (1 - \cos \theta) \right] \right. \\ &\quad \left. + E_1 \left[\frac{w}{3} D^{\alpha_a}(\tan \theta) + \frac{H}{4} D^{\alpha_a}(\cos \theta) \right] \right\} \end{aligned}$$

$$\begin{aligned}
& - \frac{Hwb\theta}{4h} \psi_a(T) \{E_0 [w \tan \theta - H(1 - \cos \theta)] \\
& + E_1 [wD^{\alpha_a}(\tan \theta) + HD^{\alpha_a}(\cos \theta)]\}
\end{aligned} \tag{11a}$$

$$\begin{aligned}
M_a &= \int_0^w \sigma(x)b(x + \frac{H}{2} \tan \theta) dx \\
&= \frac{w^2b}{h} \psi_a(T) \left\{ E_0 \left[\frac{w \tan \theta}{3} + \frac{H}{4} (1 - \cos \theta) \right] \right. \\
&+ E_1 \left[\frac{w}{3} D^{\alpha_a}(\tan \theta) - \frac{H}{4} D^{\alpha_a}(\cos \theta) \right] \left. \right\} \\
&+ \frac{Hwb\theta}{4h} \psi_a(T) \{E_0 [w \tan \theta + H(1 - \cos \theta)] \\
&+ E_1 [wD^{\alpha_a}(\tan \theta) - HD^{\alpha_a}(\cos \theta)]\}.
\end{aligned} \tag{11b}$$

3.4. Analytical Method for Structural Response

For a multi-degree-of-freedom system incorporated with viscoelastic dampers, the dynamic equilibrium equations can be written as

$$\mathbf{M}\ddot{\mathbf{X}}(t) + \mathbf{C}\dot{\mathbf{X}}(t) + \mathbf{K}\mathbf{X}(t) + \mathbf{R}(t) = \mathbf{P}(t) \tag{12}$$

where $\mathbf{M}$ and $\mathbf{C}$ are the mass matrix and damping matrix respectively, $\mathbf{K}$ is the stiffness matrix of the structure without dampers, $\mathbf{R}(t)$ is the restoring force vector induced by viscoelastic dampers, and $\mathbf{P}(t)$ is the external force vector.

To investigate the effect of connection dampers on the dynamic behaviors of a beam, the structure is meshed with beam elements while special consideration is taken into account for the connections. Assume i is the node for the beam-column connection. The relative rotation of the beam around the column is θ_i . The restoring force vector $\mathbf{R}(t)$ caused by the dampers at this connection can be written as

$$\begin{aligned}
\mathbf{R}(t) &= (\cdots F_{x_i} \ F_{y_i} \ M_{\theta_i} \ \cdots)^T = \left\{ \cdots \sum_{ii=1}^{nn} F_{s_{ii}} \ \sum_{ii=1}^{nn} F_{a_{ii}} \ \sum_{ii=1}^{nn} (M_{s_{ii}} + M_{a_{ii}}) \ \cdots \right\}^T \\
& \quad (nn = 1 \text{ or } 2)
\end{aligned} \tag{13}$$

where $F_{s_{ii}}$ is the force induced by the ii th damper at the joint i in the shear direction, $F_{a_{ii}}$ is the force induced by the ii th damper in the axial direction, $M_{s_{ii}}$ is the moment around the joint i induced by the shear restoring force of the ii th damper, $M_{a_{ii}}$ and is the moment around joint i induced by the axial restoring force of the ii th damper. The terms $F_{s_{ii}}$, $M_{s_{ii}}$, $F_{a_{ii}}$, and $M_{a_{ii}}$ are defined by equations (4), (5), (10), and (11), respectively.

The damping effect contributed by the viscoelastic dampers in the beam-column connections has been represented by $\mathbf{R}(t)$. However, the damping of the beam structure itself should also be included. Rayleigh damping is assumed for the structure as

$$\mathbf{C} = a_0 \mathbf{M} + a_1 \mathbf{K} \quad (14)$$

where parameters a_0 and a_1 are determined by

$$[a_0 \ a_1]^T = \frac{2\zeta}{\omega_1 + \omega_2} [\omega_1 \omega_2 \ 1]^T \quad (15)$$

in which ζ is the first mode damping ratio, and ω_1 and ω_2 are the first two natural frequencies of the beam structure, respectively.

A numerical method is introduced to solve the equilibrium equation in the time domain. For a degree of freedom x , the acceleration can be represented by a central difference scheme as follows (Clough and Penzien, 1993)

$$\ddot{x}(t) = \frac{1}{\delta t^2} [x(t + \delta t) - 2x(t) + x(t - \delta t)] \quad (16)$$

and velocity is written by

$$\dot{x}(t) = \frac{1}{2\delta t} [x(t + \delta t) - x(t - \delta t)]. \quad (17)$$

The fractional derivative term can be described in terms of the Γ function

$$D^\alpha x(t) = \frac{1}{\Gamma(1-\alpha)} \frac{d}{dt} \int_0^t \frac{x(\tau)}{(t-\tau)^\alpha} d\tau = \frac{1}{\Gamma(1-\alpha)} \left[\frac{x(0)}{t^\alpha} + \int_0^t \frac{\dot{x}(\tau)}{(t-\tau)^\alpha} d\tau \right]. \quad (18)$$

A change of variables in the above integral gives

$$D^\alpha x(t) = \frac{1}{\Gamma(1-\alpha)} \left[\frac{x(0)}{t^\alpha} + \int_0^t \frac{\dot{x}(t-\tau)}{\tau^\alpha} d\tau \right]. \quad (19)$$

Note that the first term is a singular term. However, since the system is assumed to be initially at rest, i.e. $x(0) = 0$, this term vanishes.

Thus, the fractional derivative term at the n th step ($t = n\delta t$) can be derived to be

$$\begin{aligned} D^\alpha x(t = n\delta t) &= \frac{1}{(\delta t)^\alpha \Gamma(2-\alpha)} \\ &\times \sum_{j=0}^{n-1} \{x[(n-j)\delta t] - x[(n-j-1)\delta t]\} [(j+1)^{1-\alpha} - j^{1-\alpha}]. \end{aligned} \quad (20)$$

With iterations being carried out for $D^\alpha x(t)$, the restoring force vector $\mathbf{R}(t)$ can be obtained in each time-step. Therefore, the equilibrium equations are solved step-by-step throughout the time-span of interest. However, this method is only conditionally stable with

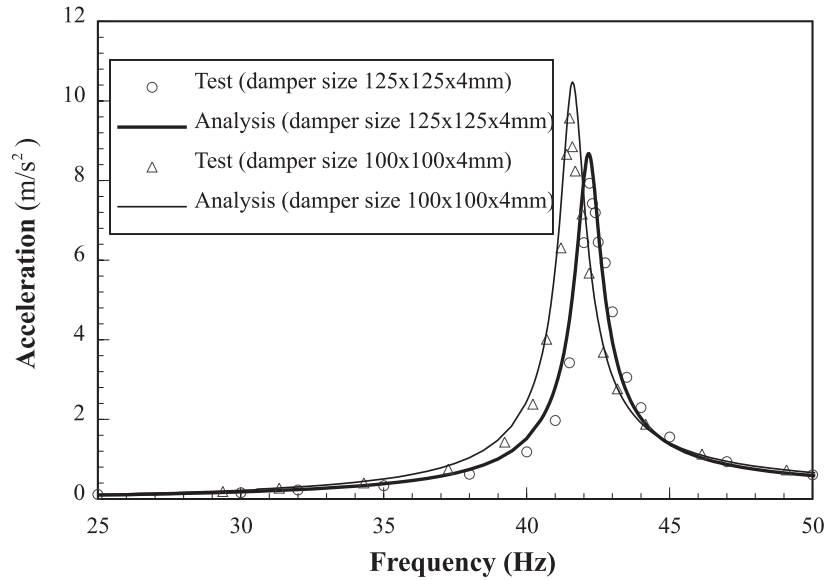


Figure 11. Comparison of acceleration peaks at section E.

a short enough time-step. The specific condition for stability (Clough and Penzien, 1993) applied in the numerical analysis is

$$\frac{\delta t}{T} \leq \frac{1}{\pi} = 0.318 \quad (21)$$

where T is the vibration period of the structure.

3.5. Verification of the Analytical Method with Experimental Results

With the developed method incorporated in finite element analysis, the tested beam structure is analyzed. Comparisons of the acceleration peaks at section E of the beam installed with dampers of various dimensions (i.e. $100 \times 100 \times 4 \text{ mm}^3$ and $125 \times 125 \times 4 \text{ mm}^3$) within the first resonant frequency region are displayed in Figure 11. It can be seen that a good agreement of the experimental and analytical results is achieved. That is to say, the analytical method can well predict the dynamic response of the beam with the proposed beam–column connections. However, some discrepancy exists between the experimental results and the analytical results. The discrepancy mainly comes from the precision of the instruments, environmental disturbance in the experimental results, and idealizations of the model in the analysis.

4. NUMERICAL STUDIES ON A LONG-SPAN BEAM

Experimental and analytical studies have shown that the response of a beam in the vertical direction can be attenuated greatly if the proposed beam–column connection is applied. The

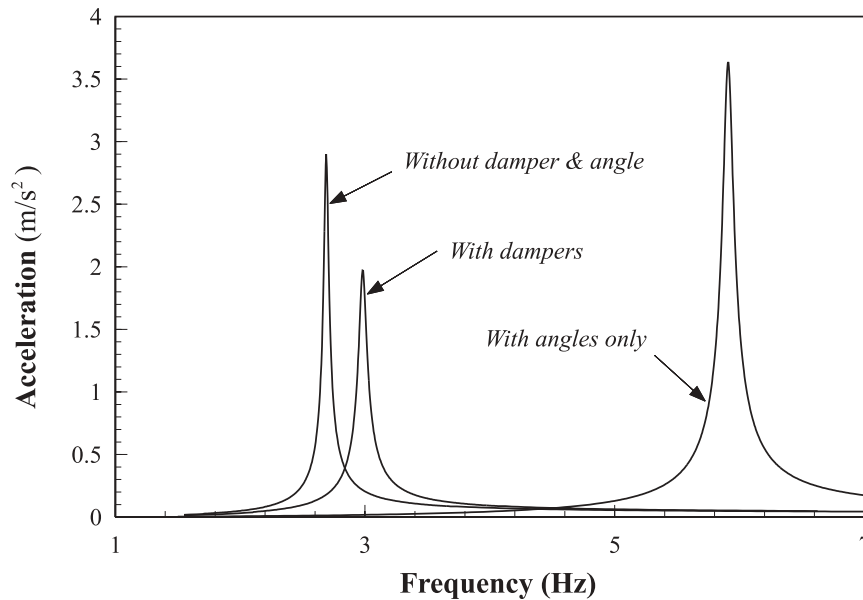


Figure 12. Acceleration peaks at section E of the long span beam within the first natural frequency region.

effectiveness in vibration control may be concerned with factors such as damper material, damper size, damper location, damper number, loading conditions, and so on. In comparison with the tested beam, the main beams used for large column-free floor systems are usually much longer, and the mass of the floor slab and other additional members should be considered also. A long-span beam similar to the main beam of the floor studied by Osborne and Ellis (1995) is investigated by numerical simulations here. An I-section beam with a height of 546 mm and a width of 406 mm, whose span is 16 m long, is selected for study. The additional mass added on the beam by secondary beams and floor is 1400 kg m^{-1} . Referring to the test results by Osborne, the first and second modal damping ratios of the beam without a damper are assumed to be 0.70%. Four DHVs with dimensions of $400 \times 400 \times 4 \text{ mm}^3$ are incorporated in the beam-column connections. To simulate the effect of human activities, a sinusoidal loading with amplitude of 600 N is imposed on the beam. The temperature is assumed to be 25°C .

Some basic works are carried out based on the above-mentioned parameters. The load is applied at the mid-span of the beam within the first natural frequency region. Acceleration peaks at the mid-span of the beam with three different beam-column connections are compared as shown in Figure 12. The resonant accelerations are 2.9, 1.97, and 3.62 m s^{-2} , respectively. With dampers incorporated, the first modal damping ratio of the beam increases from 0.70% to 1.05%. Since the second natural frequency for this long-span beam is also very low, the load is imposed at the quarter-span of the beam to investigate the effect within the second natural frequency region. Effectiveness is obvious from the acceleration peak curves at the quarter-span of the beam as illustrated in Figure 13. In the following subsections, except those to be studied, parameters are set to be the same as those in the basic works. Furthermore, studies mainly focus on the first natural frequency region.

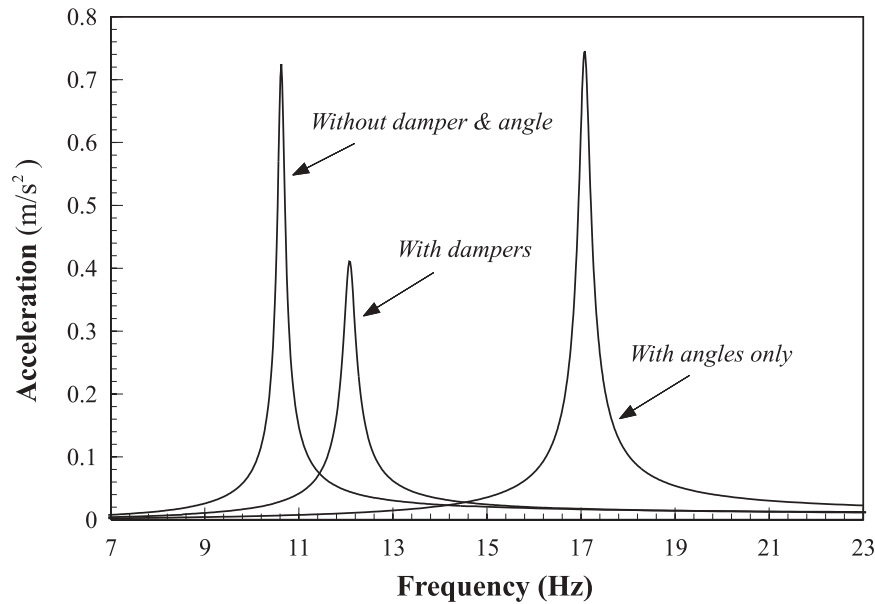


Figure 13. Acceleration peaks at section C of the long span beam within the second natural frequency region.

4.1. Variation of Damper Material

When viscoelastic dampers are to be incorporated in a structure, choosing a suitable damper material is very important. The parameter G_1 of a damper, which determines the energy-dissipation ability, is varied from 0.25 to 2.0 times the value of DHV. Curves of the acceleration peak at the mid-span of the beam are plotted in Figure 14. The resonant accelerations at the middle point of the beam and the first modal damping ratios with respect to the parameter G_1 are also summarized in Figure 15. Clearly, when G_1 increases, the resonant acceleration decreases while the first modal damping ratio increases.

4.2. Variation of Damper Size

The restoring force of a damper depends on both the damper area and the damper thickness. When the damper area increases, the restoring force increases as well. On the other hand, if a thicker damper is used, the maximum shear strain becomes smaller, which results in a decrement of the restoring force. Studies are therefore carried out on the damper thickness and damper area.

Behaviors of the beam, with damper thickness changed from 1 to 10 mm under cyclic loading at various excitation frequencies, are simulated. The resonant accelerations and the first modal damping ratios are compared in Figure 16. It can be seen that when the damper thickness increases, the resonant acceleration becomes larger while the damping ratio of the beam decreases with damper thickness nonlinearly.

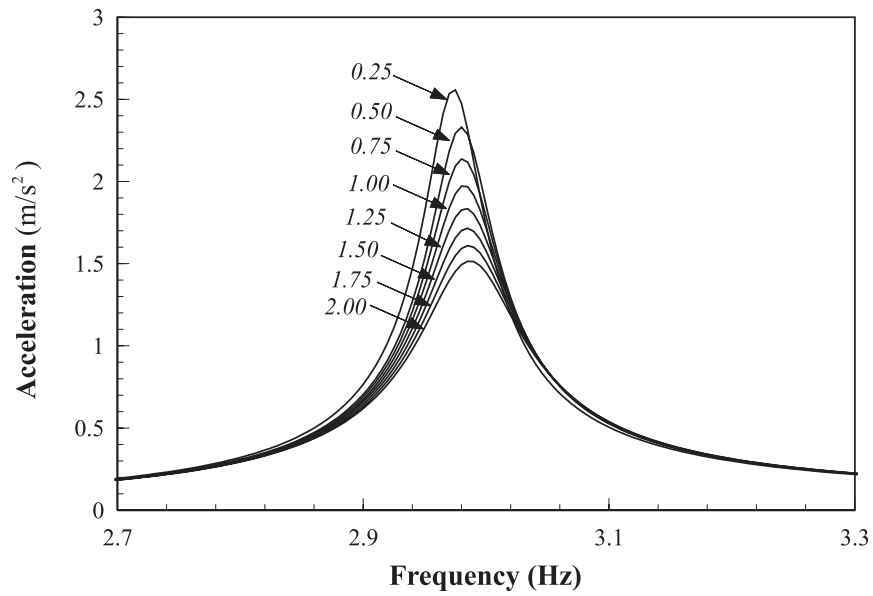


Figure 14. Acceleration peaks at the middle point of the beam with dampers of different parameter G_1 ($0.25 - 2.00 \times G_{1(DHV)}$).

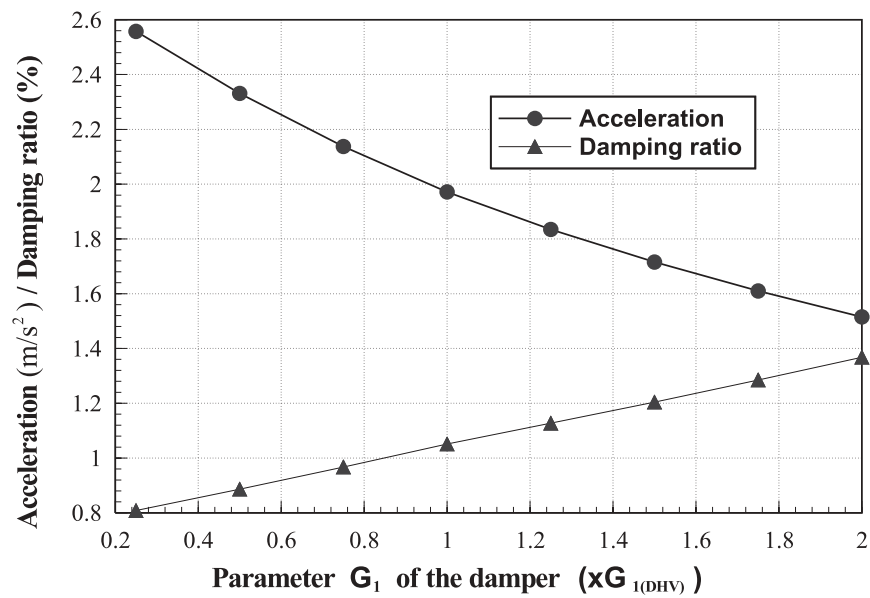


Figure 15. Resonant acceleration peaks at the middle section and the first modal damping ratios of the beam versus G_1 .

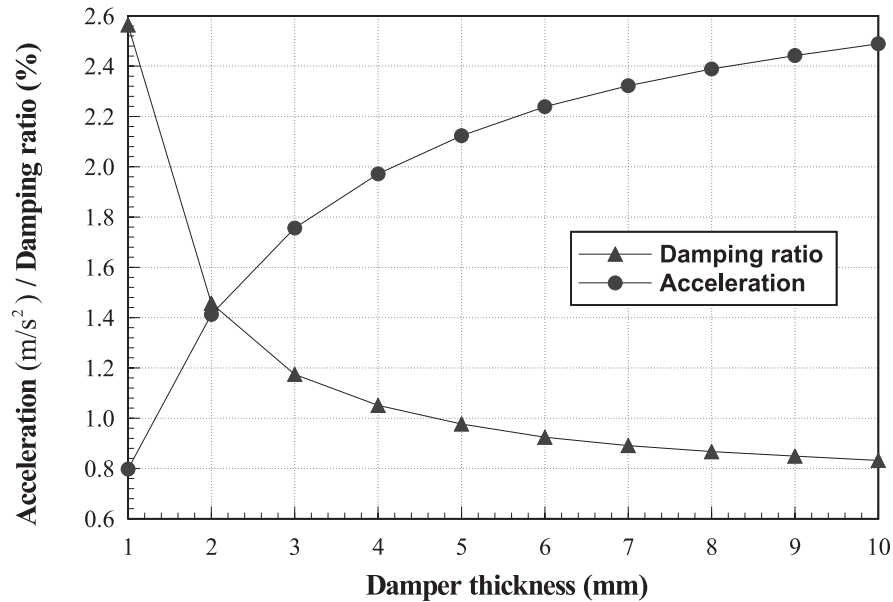


Figure 16. Resonant acceleration peaks at the middle section and the first modal damping ratio of the beam versus damper thickness.

The area of each damper is assumed to change from 0.25 to 2.00 times A_0 ($400 \times 400 \text{ mm}^2$). Resonant accelerations as well as the first modal damping ratios taken from the simulation results are shown in Figure 17. Without doubt, the resonant acceleration decreases nonlinearly with the damper area whereas the damping ratio increases nearly linearly with the damper area.

4.3. Variation of Temperature

From simulations of the beam at temperatures of 20°C , 25°C , 30°C , 35°C , and 40°C , it is obvious that the resonant responses increase greatly with temperature while the first modal damping ratio of the beam decreases. This is because when temperature increases the restoring force of a damper becomes smaller. Curves for the resonant accelerations and the first modal damping ratio under various temperatures are shown in Figure 18.

4.4. Variation of Damper Number

In general, a total of four dampers can be incorporated in the two connections of a beam. However, the damper number should be determined by the practical conditions. For example, dampers can be used only on the bottom flange of a beam on the roof of a building. Simulations of a beam with different numbers of dampers are carried out. Four cases have been considered as follows: 1, one damper at one end of the beam; 2, one damper at each end; 3, two dampers at one end and one damper at the other end; 4, two dampers at each end.

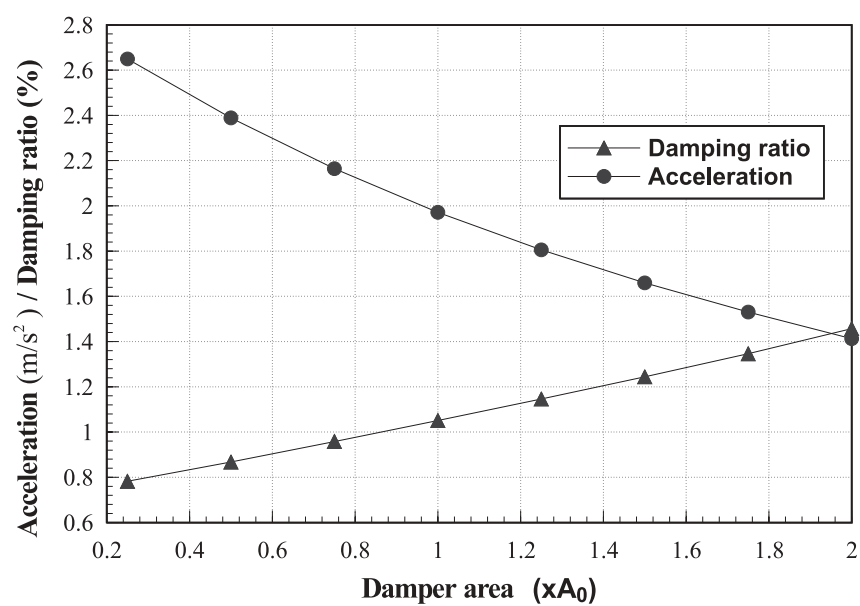


Figure 17. Resonant acceleration peaks at the middle section and the first modal damping ratio of the beam versus damper area.

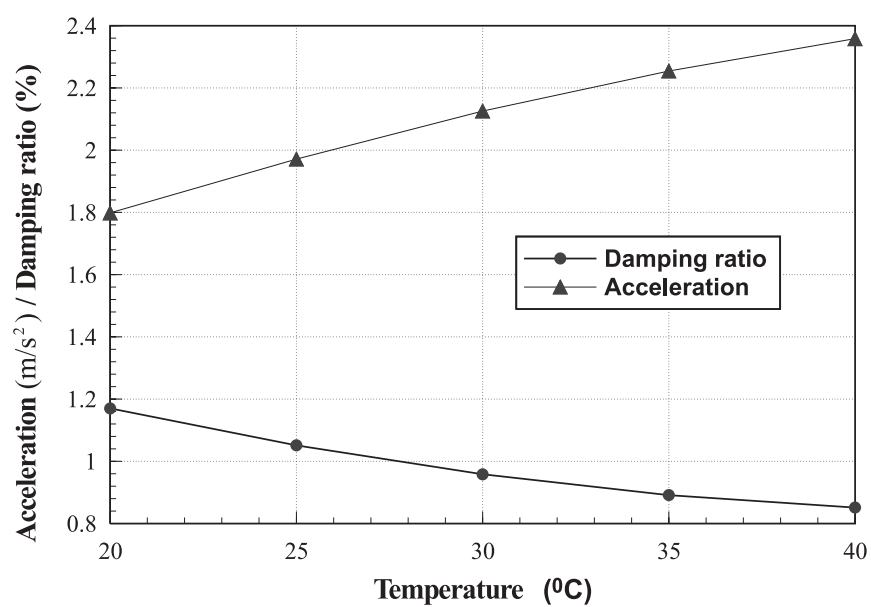


Figure 18. Resonant acceleration peaks at the middle section and the first modal damping ratio of the beam versus temperature.

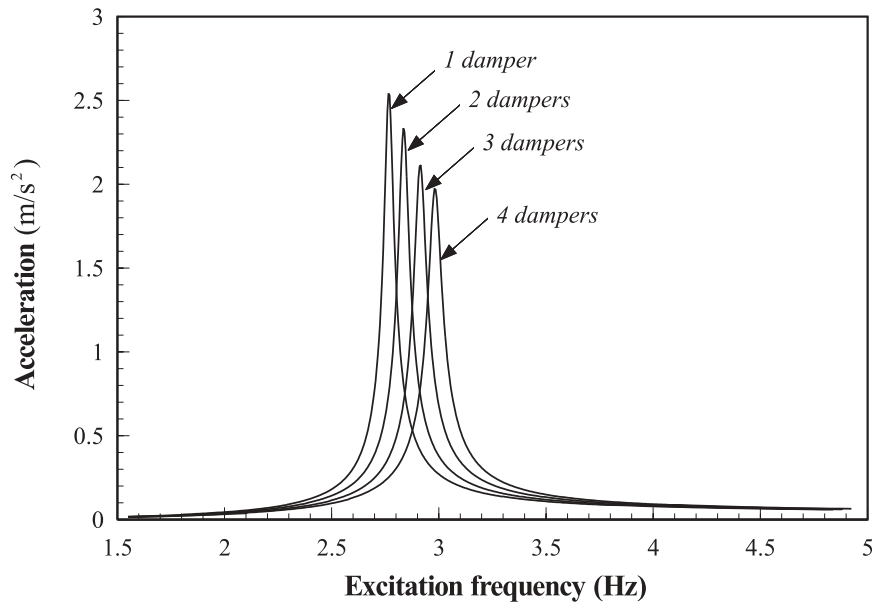


Figure 19. Acceleration peaks at the middle point of the beam with different numbers of dampers.

The curves of acceleration peak at the middle point of the beam are compared in Figure 19. The resonant frequency increases with the damper number owing to both shear stiffness and axial stiffness of dampers. The resonant accelerations together with the first modal damping ratios are summarized in Figure 20. It is obvious that better response-attenuation effect can be achieved by adding more dampers.

The influence of the damper arrangement pattern is also studied. If two dampers are incorporated in the beam–column connections, the arrangement can be set either as both two dampers on one side (case A) or as one damper on each side (case B). The acceleration peak curves at the middle point of the beam are compared as shown in Figure 21. The difference between the two cases is very minor. However, the resonant acceleration for case A is larger than that of case B, which suggests that setting one damper at each end is slightly better than setting both dampers at one end.

4.5. Variation of Loading Position and Intensity

We investigate the effectiveness in vibration control of the structure with loading acting at different positions such as 1/8, 1/4, 3/8, and 1/2 span of the beam. Compared to the beam without dampers, the maximum responses can be reduced greatly with the same ratio for all cases. This means that the effectiveness is the same no matter where the loading position is. Moreover, modal damping ratios are an internal characteristic of a structure. The first modal damping ratio of the beam remains 1.05%, no matter where the loading position is.

Behaviors of the beam under loading with amplitudes of 0.1–1 kN are also analyzed. The maximum resonant accelerations of the beam with and without dampers are presented in Figure 22. It can be seen that the ratio of the resonant acceleration at the middle of the beam

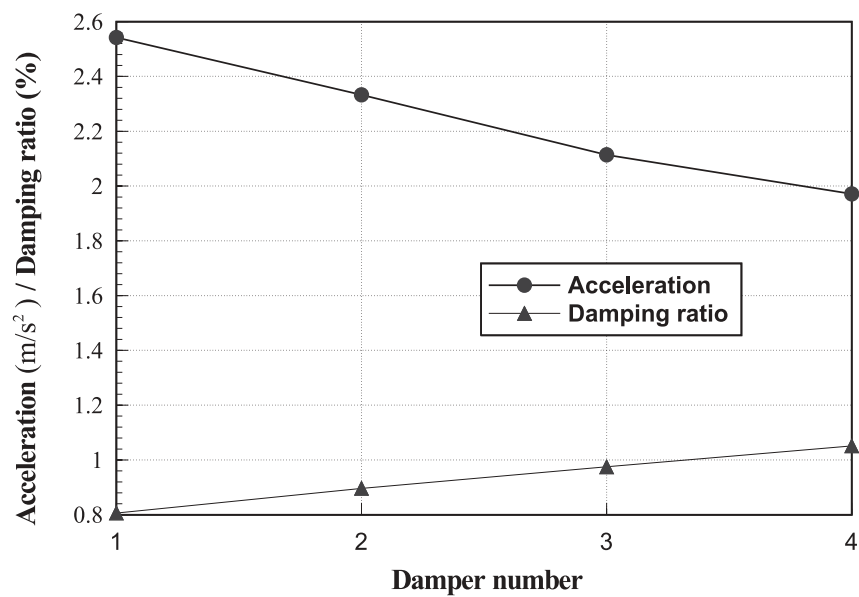


Figure 20. Resonant acceleration peaks at the middle section and the first modal damping ratio of the beam versus damper number.

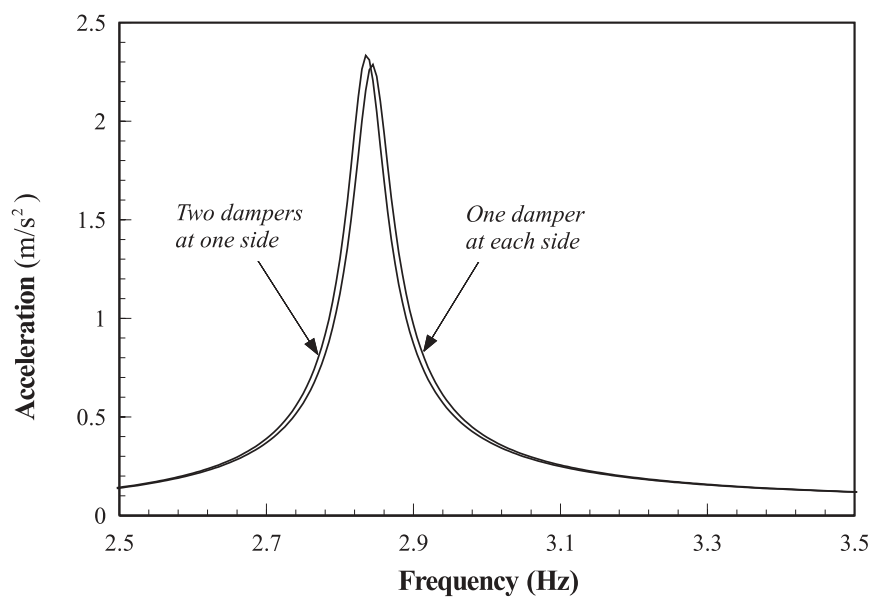


Figure 21. Acceleration peaks at the middle point of the beam with dampers of different arrangement patterns.

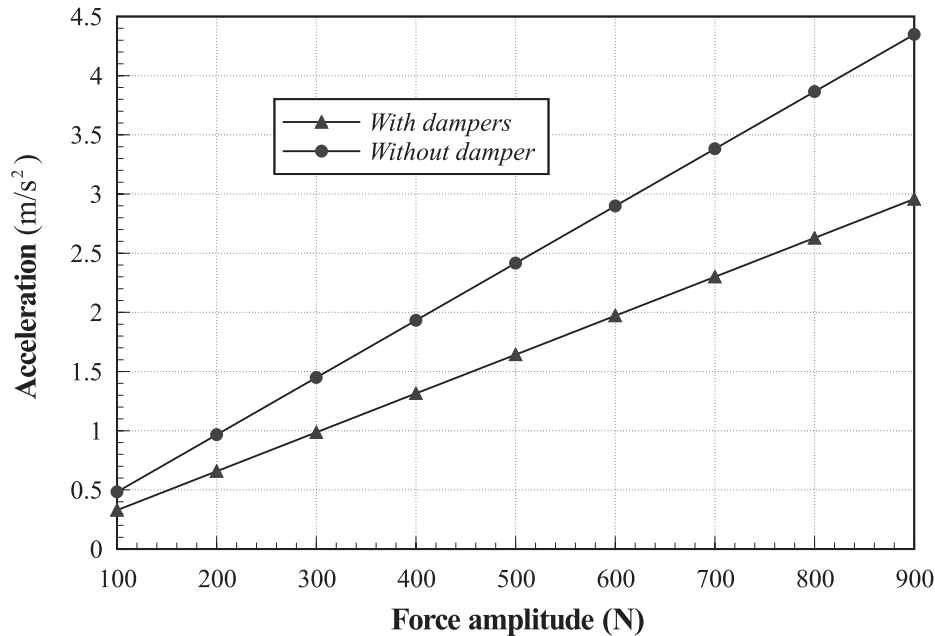


Figure 22. Resonant accelerations of the beam with and without dampers versus force amplitude.

with dampers compared to that without a damper is the same for all cases as 68%. No doubt, the first modal damping ratio of the beam also keeps the same value 1.05%.

In the frequency range of 3–6 Hz, the velocities and displacements at resonance are significant in evaluating annoyance to people. Their behaviors are therefore investigated also. The maximum resonant velocities and displacements of the beam with and without dampers under loading with amplitudes of 0.1–1 kN are compared in Figures 23 and 24, respectively. The ratio of the resonant velocity of the beam with dampers to that without a damper is 61% while the ratio of the resonant displacement is 55%. Effectiveness is obvious from these two ratios.

5. CONCLUSIONS AND DISCUSSION

A new damping device, a type of beam–column connection incorporated with viscoelastic dampers, has been designed to control the vertical vibrations of long-span structures. Great effectiveness in vertical vibration control has been demonstrated by the experimental studies. By using the improved fractional derivative model to describe the dynamic characteristics of viscoelastic dampers, an analytical method for structures with such a damping device has been proposed. Axial and shear forces in the dampers are both considered while coupling between them is ignored. This is because the effect on each other is very minor and the parameters of the fractional derivative model are very hard to identify accurately if coupling is involved. Comparisons of the experimental and analytical results for a beam structure have proved that the developed analytical method can well predict the behavior of a structure with

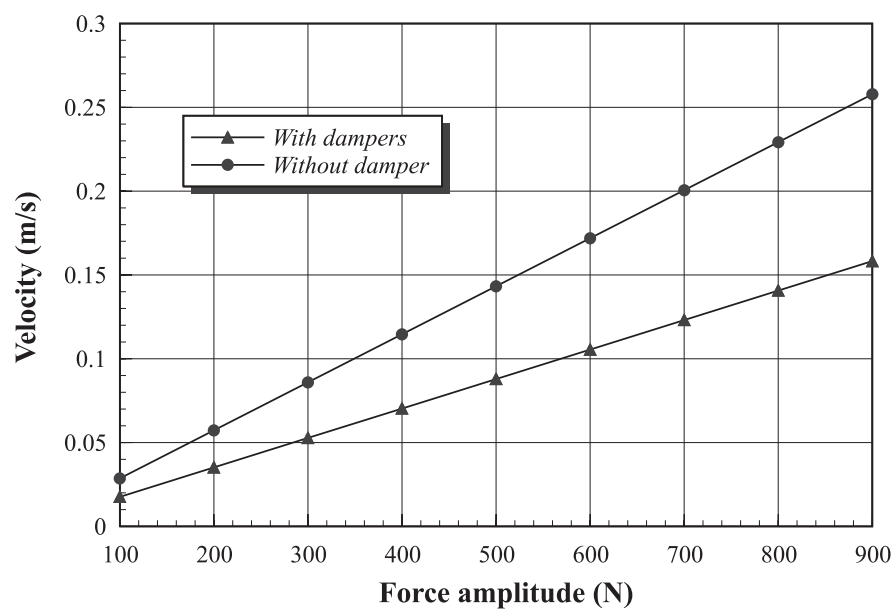


Figure 23. Resonant velocities of the beam with and without dampers versus force amplitude.

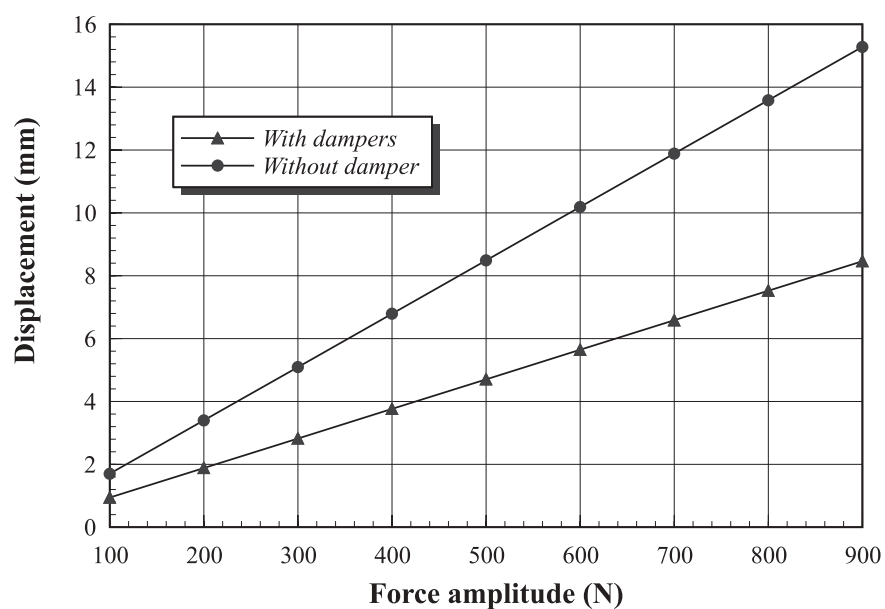


Figure 24. Resonant displacements of the beam with and without dampers versus force amplitude.

such connections. This also suggests that the coupling between axial and shear forces in the dampers is negligible for this type of application. Furthermore, guidelines for practical design of long-span structures with the proposed beam-column connections have been obtained from the parametric studies as follows.

1. It would be more effective for vibration control if the viscoelastic material being used has good energy-dissipation ability (i.e. with a high value of parameter G_1).
2. The effectiveness of vibration control can be improved by choosing proper damper dimensions such as large area and small thickness. However, if a damper is too thin, the shear strain of a damper may exceed the shear strain limit (approximately 250%) under severe conditions. Thus, the damper thickness should be determined according to assessment of loading conditions.
3. For structures in areas where temperature variation in a year is significant, dampers whose properties are less sensitive to temperature should be chosen. In addition, enough clear space should be provided for heat dissipation. Due to aging effects, maintenance may be required for the dampers after several years. The damper devices shown in Figure 2 are suggested for use. These can be taken out easily for replacement or repair.

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